



Modeling, analysis, and adaptive neural modified-backstepping control of an uncertain horizontal pendubot with double flexible joints

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ABSTRACT

Position control of an underactuated uncertain horizontal pendubot with double flexible joints (PDFJ) that has only one actuator is a new and challenging topic. This study establishes the dynamics model of the PDFJ, analyzes its characteristics, and presents an adaptive neural modified-backstepping method to achieve its position control. Through theoretical analysis, we obtain the relationship between underactuated system dynamics and the kinetic energy derivative of the two links. On this basis, an adaptive neural modified-backstepping method is proposed, which deals with the effects of parameter perturbation, model uncertainty, and external disturbance and makes the PDFJ reach the target position. This method relaxes the strict restriction of conventional backstepping methods and dynamic surface control methods that demands that the relative degree of each virtual subsystem must be one, and it can still alleviate a differential explosion problem. Finally, we demonstrate its effectiveness through simulation and experimental results.

1. Introduction

Some joints in flexible-joint (FJ) robots are made of low-stiffness or elastic materials (Sun et al., 2018). Since these kinds of robots are flexible and compliant, and they have the potential to reduce energy consumption and improve man-machine security, they have been attracting great attention and have been applied to many fields, such as manufacturing, healthcare, and aerospace (Ding et al., 2022; Kim, 2018). Some new types of robots have been developed using this kind of joint, for example, cable-driven robots (Yüksel et al., 2015), manipulators with harmonic drive gears (Trung & Iwasaki, 2022), and robots with series elastic actuators (Liu et al., 2021).

FJs in robots are divided into two kinds: active and passive, depending on whether they are directly driven by actuators. To illustrate what roles the active and passive FJs play, we first consider a basic single-link manipulator with an active first FJ. Clearly, the end load of this manipulator moves in a circle. Once there is an obstacle on the circle, the end load is inevitably affected. When we divide the single link into two links connected by a passive second FJ, the added passive FJ makes the system increase one degree of freedom, thus allowing the end load to move flexibly around the circle and avoid the obstacle. Meanwhile, this two-link manipulator makes the end load reach the same target position more compliant and safer than the single-link one. These features

make robots with passive and active FJs (RAPFJs) highly promising in some unstructured and dynamically changing workspaces. Examples of such robots include an aircraft that contained a manipulator with a passive FJ to perform tasks in unforeseen environments (Yüksel et al., 2015) and a bionic spider whose end joints of its legs were simplified to constrained passive FJs (Ming et al., 2021). But on the whole, the application fields of such robots with active and passive FJs have not been fully explored yet, and the focus of research on FJ robots has so far remained on robots with all active FJs (Luca & Book, 2008). This may be due to the great difficulty in the motion control of the systems with passive FJs.

Through analysis, the control challenges of RAPFJs come from two aspects. On the one hand, from the perspective of the dynamic model of an FJ robot, since the number of degrees-of-freedom (DOF) is greater than the number of control inputs, a RAPFJ has underactuated characteristics (Bullo & Lynch, 2001; Meng et al., 2022). Then, its underactuated equations are recognized to be nonholonomic constraints with non-integrable accelerations and velocities (Wu et al., 2019), and these equations are strongly coupled. On the other hand, due to the effect of joint flexibility, an FJ robot is more prone to vibration than a rigid-joint robot (John & Zafer, 2017). Furthermore, a RAPFJ includes two completely different types of vibration modes induced by the active

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and passive FJs, respectively. Thus, how to suppress the vibration from different sources needs to be considered in the motion control of a RAPFJ. In conclusion, these difficulties pose great challenges to the motion control of the RAPFJ.

In this paper, we try to solve the position control problem for the RAPFJ moving in the horizontal plane. Here, we take the RAPFJ with two links mentioned in the second paragraph as the control object. Since its first joint is active and the second is passive (a robot with this mechanical structure is generally called a pendubot), and both joints are flexible, we call it a pendubot with double FJs (PDFJ). Due to its concise mechanical structure and classic dynamic model with underactuated characteristics, the PDFJ is a typical RAPFJ. The aim of position control of the PDFJ is to design a control method to enable the PDFJ to track a step function with high precision so that the PDFJ achieves point-to-point control.

For a PDFJ, a key issue to achieve the goal is how to solve the motion control problem for an uncertain nonlinear system with three DOF and strong coupling using only one control input (i.e., the torque generated by the actuator at the first joint). In Madonski et al. (2020), Shi et al. (2022a), some studies on motion control of N -mass-spring systems with N -DOF ($N \geq 3$) were developed. But this system is usually simplified to a linear model before designing the controller. When we use a classical Lagrangian modeling method to establish the dynamic model, we find that the system is not strict feedback (Duan, 2021). This makes motion control of the PDFJ very difficult, and even just implementing its position control is a great challenge. At present, there have been few effective control strategies for the PDFJ. However, there were some studies for systems with similar underactuated characteristics. For vertical manipulators with passive rigid joints, some energy-based swing-up controllers were designed (Lai et al., 2000; Liu & Xin, 2017). For a three-link passive-passive-active manipulator, a damping controller based on the Lyapunov method was developed by Chen and Goodwine (2019), and a tracking control method with trajectory optimization was presented by Wang et al. (2021a). For a helicopter with three DOF, an adaptive neural-network backstepping method was designed by Yang and Zheng (2020). Although these methods may be difficult to apply to the PDFJ due to the differences in system dynamics and control objectives, we draw an important conclusion by analyzing why these methods are able to perform motion control. Namely, these methods exploited the approximate linear controllability of the systems at their equivalence points. It points to a path for the motion control research of the PDFJ.

To achieve motion control and vibration suppression of FJ robots, some people used dynamic surface control methods (Li et al., 2012), backstepping methods with command filters (Dian et al., 2019), and other backstepping-based methods (Li et al., 2022; Shi et al., 2022b; Sun et al., 2021). On this basis, some adaptive learning techniques (Li et al., 2022, 2012; Sun et al., 2021) were employed to estimate unknown disturbances. Some people designed sliding mode controllers, and the resulting chattering was reduced by adding some approximators (Rsetam et al., 2020; Xiong et al., 2019). In Yan et al. (2021), an equivalent-input-disturbance-based method was presented for the tracking control of an FJM with a single link. In Zhang et al. (2015), the global stabilization of a three-link horizontal manipulator with a passive first FJ was addressed, but this study cannot achieve position control. In addition, some people employed time-scale expansion-based methods (Zhang & Li, 2017), passivity control methods (Keppler et al., 2016), and coupling-torque-compensation-based methods (Trung & Iwasaki, 2022) to suppress vibration. Overall, these methods were designed for robots with all active FJs that are strict feedback and did not have the ability to directly suppress the vibration induced by passive FJs. Thus, they cannot be directly applied to the PDFJ. Nevertheless, the backstepping method has a distinctive design process, which is to decompose a high-order system into several virtual subsystems with a relative degree one for controller design (Kokotovic et al., 1992). Considering that the PDFJ is a higher-order system, using the backstepping method for the PDFJ is feasible by splitting the system into

four virtual subsystems, but the increase in the number of subsystems and the strongly coupled underactuated equations lead to a differential explosion problem (Cheng et al., 2021) and then make controller design very complex. Hence, a modified-backstepping method with the ability to overcome the above drawbacks is worth investigating.

In addition, the position control of the PDFJ needs to take into account some uncertainties in the dynamic model, such as parameter perturbation and external disturbance, so as to ensure high control performance. In this regard, some people designed active disturbance rejection control methods to estimate and compensate for disturbances, in which the observer and the controller are usually designed based on an ideal chained integrator model (Chen et al., 2016; Zhang et al., 2016). However, building such an ideal model is quite difficult for the PDFJ because the system (with single input) has a strong coupling. Some studies used neural networks (NNs) to approximate uncertainties and then presented adaptive NN-based control methods (Li et al., 2019; Li & Li, 2018; Liu et al., 2023b). Since the weights of the NN can be updated online by designing an adaptive control law, the NN usually has a good approximation capability.

Inspired by the above analysis, this paper presents an adaptive neural modified-backstepping control method for the position control of the PDFJ. First, we establish the dynamic model of the PDFJ with uncertainties, which includes two parts: actuated and underactuated equations. For the convenience of controller design, we rewrite the underactuated equations to be a dynamic model of the rigid-joint manipulator with virtual torque. Then, we find the characteristic that the approximate linear model of the PDFJ is controllable at its equivalent points. Moreover, we obtain the relationship between the virtual torque and the kinetic energy derivative of the two links. Based on the relationship, we propose an adaptive neural modified-backstepping method for the PDFJ. In the controller design, we design NN approximators to deal with the effects of parameter perturbation, model uncertainty, and some terms that are difficult to calculate, and we employ virtual damping to tackle external disturbance. It is worth mentioning that we relax the restriction of conventional backstepping methods and dynamic surface control methods that demands that the relative degree of any virtual subsystem must be one (Kokotovic et al., 1992), and alleviate the differential explosion problem existing in the conventional backstepping methods by systematic design. Finally, the stability of the closed-loop system is proved, and the validity of the proposed method is verified by experimental results.

The contributions and advantages of this paper are summarized as follows.

(1) This paper revealed two characteristics of the PDFJ, regarding the controllability of its approximate linear model and the relationship between underactuated system dynamics and the kinetic energy derivative of the two links, which are helpful for the research on motion control of the PDFJ.

(2) This study designed an adaptive neural modified-backstepping method to achieve position control of the uncertain horizontal pendubot with double flexible joints. The method combines the neural network approximators and virtual damping to overcome some uncertainties and is a robust control method.

(3) The presented method relaxes the restriction of conventional backstepping methods and dynamic surface control methods, demanding that the relative degree of any virtual subsystem must be one, and it still has the ability to alleviate the differential explosion problem via systematic design.

In addition, some symbols are noticed for the rest of this paper. $\|\Xi\|$ denotes the norm of the matrix Ξ , Ξ^{-1} denotes the inverse of Ξ , t is time, $\exp(y)$ is the exponential function of y , and $y^{(r)} = \frac{d^r}{dt^r} y$, where $r > 0$ is an integer.

2. Modeling and preparations

In this section, the dynamic model of the PDFJ is established. For the convenience of controller design, we present or derive some theoretical analysis of the dynamic model and give some preliminary knowledge.

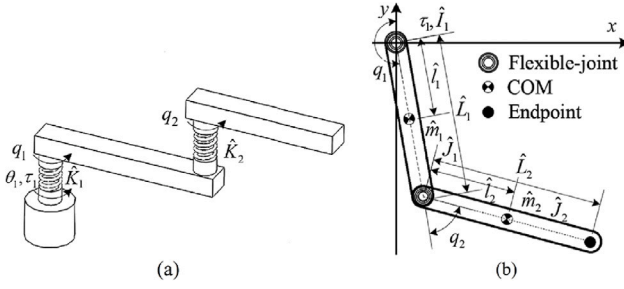


Fig. 1. Space model (a) and physical structure (b) of the PDFJ.

2.1. Modeling of the dynamics

As depicted in the space model of the PDFJ (Fig. 1(a)), the PDFJ is driven by only one available actuator in the first link that generates the control torque τ_1 . q_1 is the first link angle going through the flexible first joint. q_2 is the second link angle connecting with the end of the first link. θ_1 is the actuator angle influenced by the actuator directly. There is no actuator in the second link. Fig. 1(b) shows the physical structure of the PDFJ, where $\hat{I}_1$ denotes the actuator inertia. For the i th link ($i = 1, 2$), $\hat{m}_i$, $\hat{K}_i$, $\hat{J}_i$, $\hat{L}_i$, and $\hat{l}_i$ are its mass, joint flexibility, link inertia with respect to its center of mass (COM), length, and the distance from its joint to its COM. Due to measurement deviation or aged deterioration, the PDFJ may have parameter perturbation in practice. We define I_1 , m_i , K_i , J_i , L_i , and l_i to be the nominal values of the actual physical parameters $\hat{I}_1$, $\hat{m}_i$, $\hat{K}_i$, $\hat{J}_i$, $\hat{L}_i$, and $\hat{l}_i$. There exist the following relationships between the actual physical parameters and their nominal physical parameters: $\hat{I}_1 = (1 + k_{I_1})I_1$, $\hat{m}_i = (1 + k_{m_i})m_i$, $\hat{K}_i = (1 + k_{K_i})K_i$, $\hat{J}_i = (1 + k_{J_i})J_i$, $\hat{L}_i = (1 + k_{L_i})L_i$, and $\hat{l}_i = (1 + k_{l_i})l_i$ ($i = 1, 2$), where k_{I_1} , k_{m_i} , k_{K_i} , k_{J_i} , k_{L_i} , and k_{l_i} , which are greater than -100% and are assumed to be bounded, denote the perturbed percentages of the physical parameters.

Let $q = [q_1, q_2]^T$, and let the position vector of the COM of the i th link be Z_i ($i = 1, 2$). According to Fig. 1(b), Z_i satisfies

$$\begin{aligned} Z_1 &= [\hat{l}_1 \sin q_1, \hat{l}_1 \cos q_1] \\ Z_2 &= [\hat{L}_1 \sin q_1 + \hat{l}_1 \sin(q_1 + q_2), \hat{L}_1 \cos q_1 + \hat{l}_1 \cos(q_1 + q_2)] \end{aligned} \quad (1)$$

Based on (1), the kinetic energy of the actuator E_{k1} , the kinetic energy of the two links E_{k2} , and the elastic potential energy of the system E_p are calculated to be

$$\begin{aligned} E_{k1} &= \hat{I}_1 \dot{\theta}_1^2 / 2 \\ E_{k2} &= [\hat{m}_1 \dot{Z}_1^T \dot{Z}_1 + \hat{J}_1 \dot{q}_1^2 + \hat{m}_2 \dot{Z}_2^T \dot{Z}_2 + \hat{J}_2 (\dot{q}_1 + \dot{q}_2)^2] / 2 \\ E_p &= [\hat{K}_1 (q_1 - \theta_1)^2 + \hat{K}_2 q_2^2] / 2. \end{aligned} \quad (2)$$

Then, we define the Lagrangian function of the PDFJ to be

$$E_l = E_{k1} + E_{k2} - E_p \quad (3)$$

Employing the Lagrangian modeling method, we describe the dynamic model of the PDFJ as

$$\begin{aligned} \frac{d}{dt} \left[\frac{\partial E_l}{\partial \dot{\theta}_1} \right] - \frac{\partial E_l}{\partial \theta_1} &= \tau_1 - \Psi_\theta(q_1, \theta_1, \dot{\theta}_1, t) \\ \frac{d}{dt} \left[\frac{\partial E_l}{\partial \dot{q}} \right] - \frac{\partial E_l}{\partial q} &= -\Psi_q(\dot{q}) \end{aligned} \quad (4)$$

where $\Psi_\theta(q_1, \theta_1, \dot{\theta}_1, t)$ denotes the uncertainties vector of the actuator, and $\Psi_q(\dot{q})$ denotes the uncertainties vector of the links. The dynamic model (4) of the PDFJ is further arranged as

$$\hat{I}_1 \ddot{\theta}_1 - \hat{K}_1 \cdot (q_1 - \theta_1) = \tau_1 - \Psi_\theta(q_1, \theta_1, \dot{\theta}_1, t) \quad (5a)$$

$$\hat{M}(q_2) \ddot{q} + \hat{C}(q, \dot{q}) \dot{q} + \hat{P}(q, \theta_1) = -\Psi_q(\dot{q}) \quad (5b)$$

where $\hat{M}(q_2)$ denotes the inertia matrix, $\hat{C}(q, \dot{q}) \dot{q}$ denotes the vector of centripetal forces and Coriolis forces, and $\hat{P}(q, \theta_1)$ denotes the elastic force vector. According to the physical structure of the PDFJ, $\Psi_q(\dot{q})$ is related to link and flexible joint, and it is composed of both viscous friction and coulomb friction. $\Psi_\theta(q_1, \theta_1, \dot{\theta}_1, t)$ is related to flexible joint and actuator, and the actuator easily brings some other uncertainties except frictions.

In the dynamic model (5), $\hat{M}(q_2)$, $\hat{C}(q, \dot{q}) \dot{q}$, $\hat{P}(q, \theta_1)$, $\Psi_\theta(q_1, \theta_1, \dot{\theta}_1, t)$, and $\Psi_q(\dot{q})$ are concretely expressed as

$$\begin{aligned} \hat{M}(q_2) &= \begin{bmatrix} a_1 + a_2 + 2a_3 \cos(q_2) & a_2 + a_3 \cos(q_2) \\ a_2 + a_3 \cos(q_2) & a_2 \end{bmatrix} \\ \hat{C}(q, \dot{q}) \dot{q} &= a_3 \sin(q_2) [-\dot{q}_2^2 - 2\dot{q}_1 \dot{q}_2, \dot{q}_1^2]^T \\ \hat{P}(q, \theta_1) &= [\hat{K}_1 \cdot (q_1 - \theta_1), \hat{K}_2 q_2]^T \\ \Psi_\theta(q_1, \theta_1, \dot{\theta}_1, t) &= f_0(q_1, \theta_1, \dot{\theta}_1) + \sigma_0(t) \\ \Psi_q(\dot{q}) &= [f_1(\dot{q}_1), f_2(\dot{q}_2)]^T \end{aligned} \quad (6)$$

in which

$$\begin{aligned} a_1 &= \hat{J}_1 + \hat{m}_1 \hat{l}_1^2 + \hat{m}_2 \hat{L}_1^2, \quad a_2 = \hat{J}_2 + \hat{m}_2 \hat{l}_2^2, \quad a_3 = \hat{m}_2 \hat{L}_1 \hat{l}_2 \\ f_1(\dot{q}_1) &= \rho_{v1} \dot{q}_1 + \rho_{c1} \text{sgn}(\dot{q}_1), \quad f_2(\dot{q}_2) = \rho_{v2} \dot{q}_2 + \rho_{c2} \text{sgn}(\dot{q}_2) \end{aligned} \quad (7)$$

$f_0(q_1, \theta_1, \dot{\theta}_1)$ denotes the unknown model uncertainty, including viscous friction and coulomb friction of the actuator, and $\sigma_0(t)$ denotes the external disturbance. For the i th link, $f_i(\dot{q}_i)$ is the friction, and ρ_{vi} and ρ_{ci} are the unknown coefficients of viscous friction and coulomb friction, respectively ($i = 1, 2$).

Assumption 1. For all $t > 0$, there exists a positive constant σ_m such that $\sigma_0(t)$ satisfies $|\sigma_0(t)| \leq \sigma_m$.

2.2. Control objective and characteristics analysis

Define q_{1d} as the target angle of the first link of the PDFJ. Because the first link and the second link can be regarded as being connected only by a flexible joint, the target angle of the second link is $q_{2d} = 0$. Thus, the position control objective in this paper is to design a controller to make q_1 reach its target angle and suppress the vibration of the two links. That is, we should design an appropriate τ_1 to make $q_1 \rightarrow q_{1d}$ and $q_2 \rightarrow 0$ as accurate as possible.

Let $\bar{x} = [\theta_1, q_1, q_2, \dot{\theta}_1, \dot{q}_1, \dot{q}_2]^T$. According to the physical structure of the PDFJ and the dynamic model (5), we know the equivalent point of the PDFJ is $\bar{x}^\circ = [\theta_1^\circ, q_1^\circ, 0, 0, 0, 0]^T$, where $\theta_1^\circ = q_1^\circ$, and q_1° can be any angle on the horizontal plane. From Section 1, we know that the approximate linear models of some underactuated systems with multiple underactuated states are controllable at their equivalent points. Is also the approximate linear model of the PDFJ controllable at its equivalent point? Next, we will study this problem.

Here, two properties with respect to the flexible-joint manipulator (Ghorbel et al., 1989) are listed below.

Property 1. $\hat{M}(q_2)$ is positive definite and symmetric.

Property 2. $\forall q, \dot{q} \in \mathbb{R}^2$: $\dot{q}^T [\hat{M}(q_2)/2 - \hat{C}(q, \dot{q})] \dot{q} = 0$.

Based on (5)–(7) and Property 1, the approximate linear model of the PDFJ at the equivalent point is

$$\dot{\bar{x}} = F(\bar{x} - \bar{x}^\circ) + G\tau_1 \quad (8)$$

where

$$\begin{aligned} F &= \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ \delta_{41} & \delta_{42} & 0 & \delta_{44} & 0 & 0 \\ \delta_{51} & \delta_{52} & \delta_{53} & 0 & \delta_{55} & \delta_{56} \\ \delta_{61} & \delta_{62} & \delta_{63} & 0 & \delta_{65} & \delta_{66} \end{bmatrix} \\ G &= [0, 0, 0, 1/\hat{I}_1, 0, 0]^T \end{aligned} \quad (9)$$

For the matrix F , we obtain that

$$\begin{aligned}\delta_{41} &= -(\hat{K}_1 + \tilde{f}_{01})/\hat{I}_1, \delta_{42} = (\hat{K}_1 - \tilde{f}_{02})/\hat{I}_1 \\ \delta_{44} &= -\tilde{f}_{04}/\hat{I}_1, \delta_{51} = N_{11}\hat{K}_1, \delta_{52} = -\delta_{51}, \delta_{53} = -N_{12}\hat{K}_2 \\ \delta_{55} &= -N_{11}\tilde{f}_{15}, \delta_{56} = -N_{12}\tilde{f}_{16}, \delta_{61} = N_{21}\hat{K}_1, \delta_{62} = -\delta_{61} \\ \delta_{63} &= -N_{22}\hat{K}_2, \delta_{65} = -N_{21}\tilde{f}_{15}, \delta_{66} = -N_{22}\tilde{f}_{16} \\ \tilde{f}_{01} &= \left. \frac{\partial \Psi_\theta(q_1, \theta_1, \dot{\theta}_1)}{\partial \theta_1} \right|_{\tilde{x}=\tilde{x}^0}, \tilde{f}_{02} = \left. \frac{\partial \Psi_\theta(q_1, \theta_1, \dot{\theta}_1)}{\partial q_1} \right|_{\tilde{x}=\tilde{x}^0} \\ \tilde{f}_{04} &= \left. \frac{\partial \Psi_\theta(q_1, \theta_1, \dot{\theta}_1)}{\partial \theta_1} \right|_{\tilde{x}=\tilde{x}^0}, \tilde{f}_{15} = \left. \frac{\partial f_1(q_1)}{\partial q_1} \right|_{q_1=0}, \tilde{f}_{16} = \left. \frac{\partial f_2(q_2)}{\partial q_2} \right|_{q_1=0}\end{aligned}\quad (10)$$

and N_{11} , N_{12} , N_{21} , and N_{22} satisfy that

$$\begin{aligned}\hat{M}^{-1}(q_2)|_{q_2=0} &= \begin{bmatrix} N_{11} & N_{12} \\ N_{21} & N_{22} \end{bmatrix}, N_{11} = -\frac{a_2}{a_1 a_2 - a_3^2} \\ N_{12} &= N_{21} = \frac{a_2 + a_3}{a_1 a_2 - a_3^2}, N_{22} = -\frac{a_1 + a_2 + 2a_3}{a_1 a_2 - a_3^2}, a_1 a_2 > a_3^2\end{aligned}\quad (11)$$

We obtain that the controllability matrix of (8) is $C_{ep} = [G, FG, F^2G, F^3G, F^4G, F^5G]$, and its determinant $|C_{ep}|$ is

$$|C_{ep}| = \hat{K}_1^4 \hat{K}_2^2 N_{21}^2 (N_{11} N_{22} - N_{12} N_{21})^2 / \hat{I}_1^6 \quad (12)$$

Eq. (12) shows that the controllability of the approximate linear model of the PDFJ at the equivalent point is independent of unknown model uncertainties containing frictions. According to Property 1, we know that the determinant $\det(\hat{M}^{-1}(q_2)|_{q_2=0}) > 0$. Applying the result to (11) produces $N_{11} N_{22} - N_{12} N_{21} > 0$. Hence, it is clear that $|C_{ep}| > 0$, which indicates that unlike the HP with rigid joints, the approximate linear model of the PDFJ is controllable at the equivalent point. It provides a theoretical basis for our controller design in the next section.

Besides, based on (5) and Properties 1 and 2, the time derivative of E_{k2} satisfies that

$$\begin{aligned}\dot{E}_{k2} &= \frac{d}{dt} [\dot{q}^T \hat{M}(q_2) \dot{q} / 2] \\ &= [\dot{q}^T \hat{M}(q_2) \ddot{q} + \dot{q}^T \dot{\hat{M}}(q_2) \dot{q} + \dot{q}^T \hat{M}(q_2) \ddot{q}] / 2 \\ &= \dot{q}^T [-\hat{P}(q, \theta_1) - \Psi_q(\dot{q})] + \dot{q}^T [\dot{\hat{M}}(q_2) - 2\hat{C}(q, \dot{q})] \dot{q} / 2 \\ &= \dot{q}^T [-\hat{P}(q, \theta_1) - \Psi_q(\dot{q})]\end{aligned}\quad (13)$$

In (13), we let

$$\tau_v = \dot{q}^T [-\hat{P}(q, \theta_1) - \Psi_q(\dot{q})] \quad (14)$$

Clearly, τ_v is the underactuated system dynamics from (5b). Therefore, the relationship (13) connects the underactuated system dynamics τ_v with the kinetic energy derivative of the two links $\dot{E}_{k2}$ directly.

2.3. Preliminary knowledge

In this paper, we employ radial basis function (RBF) NN approximators to approximate some continuous functions with respect to states. Here, the unknown function to be approximated is written to be $\chi_j(v_j)$, where $v_j \in \mathbb{R}^{\ell_j}$ is a scalar or a column vector with respect to ℓ_j inputs, and $j = 1, 2, \dots, n$ represents the sequence number of n unknown functions to be approximated.

According to the research in Ge et al. (2013), Sanner and Slotine (1992), RBF NN has the ability to approximate the unknown continuous function $\chi_j(v_j)$ over a sufficiently large compact set $\Omega_j \in \mathbb{R}^{\ell_j}$, and $\chi_j(v_j)$ is approximated to be

$$\chi_j(v_j) = W_j^* T G_j(v_j) + \varepsilon_j, \forall v_j \in \Omega_j \quad (15)$$

where $W_j^* \in \mathbb{R}^{\theta_j}$ denotes an optimal weight matrix, θ_j is a positive integer, ε_j is the bounded approximate error of the NN, and $G_j(v_j) = [g_{j1}(v_j), g_{j2}(v_j), \dots, g_{j\theta_j}(v_j)]^T \in \mathbb{R}^{\theta_j}$ represents the RBF vector with θ_j neurons ($j = 1, 2, \dots, n$). It has been proven that there exists a controller with enough neurons that enables all the signals in the closed loop to remain bounded as long as v_j remains within a prefixed compact set

Ω_j , where Ω_j can be chosen as large as desired (Ge & Wang, 2004). For the RBF $g_{jr_j}(v_j)$ of the r_j th neuron ($r_j = 1, 2, \dots, \theta_j$; $j = 1, 2, \dots, n$), we choose it as the following Gaussian function:

$$g_{jr_j}(v_j) = \exp \left[-(v_j - \eta_{jr_j})^T (v_j - \eta_{jr_j}) / (2b_{jr_j}^2) \right] \quad (16)$$

in which $\eta_{jr_j} \in \mathbb{R}^{\ell_j}$ denotes the center coordinate vector of v_j , and b_{jr_j} denotes the width of the Gaussian function. For all neurons in NN, we define $\eta_j = [\eta_{j1}, \eta_{j2}, \dots, \eta_{j\theta_j}]^T$ ($r_j = 1, 2, \dots, \theta_j$; $j = 1, 2, \dots, n$).

Then, based on (15), the approximate vector of $\chi_j(v_j)$ is designed to be

$$\hat{\chi}_j(v_j) = W_j^T G_j(v_j) \quad (17)$$

where $W_j = [w_{j1}, w_{j2}, \dots, w_{j\theta_j}]^T \in \mathbb{R}^{\theta_j}$ is an adaptive weight matrix ($j = 1, 2, \dots, n$), and $w_{jr_j}(0)$ in $W_j(0)$ is set to be bounded. The error vector between W_j^* and W_j is expressed as

$$\tilde{W}_j = W_j^* - W_j. \quad (18)$$

Assumption 2. Without a loss of generality, there exists an unknown boundary $0 < \varepsilon_{jM} < \infty$ such that $|\varepsilon_j| \leq \varepsilon_{jM}$ ($j = 1, 2, \dots, n$).

In addition, for the convenience of the subsequent analysis, some properties concerning the hyperbolic tangent function $\tanh(x)$ and its time derivatives are given in advance:

$$\begin{aligned}\tanh(x) &= \frac{e^x - e^{-x}}{e^x + e^{-x}}, x \in (-\infty, \infty) \\ -1 < y = \tanh(x) < 1, 0 &\leq \dot{y} = \text{sech}^2(x) \leq 1 \\ -1 < \ddot{y} &= -2 \tanh(x) \text{sech}^2(x) < 1\end{aligned}\quad (19)$$

where $\tanh(x)$ converges smoothly to -1 and 1 as $x \rightarrow -\infty$ and $x \rightarrow \infty$, respectively, and $\text{sech}(x)$ denotes the hyperbolic secant function.

3. Adaptive neural modified-backstepping method

This section presents an adaptive neural modified-backstepping method to achieve the control objective. To enhance the readability of the whole paper, here, we show the idea of the modified-backstepping method and all the elements of the adaptive neural modified-backstepping controller in advance. The concrete design procedure and the selection principles of the design parameters will be presented later with stability analysis in the second subsection.

3.1. Adaptive neural modified-backstepping controller

As a recursive design method, the backstepping method can be applied for nonlinear system control. The basic idea of conventional backstepping methods (Kokotovic, 1992; Kokotovic et al., 1992) is to split the whole system into some virtual subsystems with relative degree one and to design corresponding Lyapunov function and virtual controller for each virtual system, finishing with the controller as the last virtual controller. In addition, some backstepping-based methods such as the common dynamic surface control methods (Jafari et al., 2019; Wang et al., 2021b; Zhang et al., 2022) and the command filtering control methods (Liu et al., 2023a; Rashad et al., 2016; Yu et al., 2018) have the same basic idea in requiring the relative degree of any virtual subsystem is one. Meanwhile, these methods require that the controlled systems are strict feedback (Duan, 2021).

However, the dynamic model (5) is not a strict-feedback system. Letting $\hat{x} = [x_1, x_2, x_3]^T = [q_2, q_1, \theta_1]^T$, we take the example of the PDFJ with the form of the following second-order system:

$$\begin{aligned}\ddot{x}_1 &= f_{x1}(x_1, \dot{x}_1, \dot{x}_2, x_3) + g_1(x_1)x_2 \\ \ddot{x}_2 &= f_{x2}(x_1, \dot{x}_1, x_2, \dot{x}_2) + g_2(x_1)x_3 \\ \ddot{x}_3 &= f_{x3}(x_2, x_3, \dot{x}_3) + g_3\tau_1 - \sigma_0(t)/\hat{I}_1\end{aligned}\quad (20)$$

where

$$\begin{aligned} f_{x1} &= \xi_2 \Gamma(x_1, \dot{x}_1, \dot{x}_2) - \xi_2 \hat{M}^{-1}(x_1) [-\hat{K}_1 x_3, \hat{K}_2 x_1]^T \\ f_{x2} &= \xi_1 \Gamma(x_1, \dot{x}_1, \dot{x}_2) - \xi_1 \hat{M}^{-1}(x_1) [\hat{K}_1 x_2, \hat{K}_2 x_1]^T \\ f_{x3} &= [\hat{K}_1(x_2 - x_3) - f_0(x_2, x_2, \dot{x}_2)] / \hat{I}_1, \quad g_3 = 1 / \hat{I}_1 \\ g_1(x_1) &= \frac{\hat{K}_1(a_2 + a_3 \cos(x_1))}{a_1 a_2 - a_3^2 \cos^2(x_1)}, \quad g_2(x_1) = \frac{\hat{K}_1 a_2}{a_1 a_2 - a_3^2 \cos^2(x_1)} \end{aligned} \quad (21)$$

where $\xi_1 = [1, 0]$, $\xi_2 = [0, 1]$, and $\Gamma = \hat{M}^{-1}(x_1)[a_3 \sin(x_1) (\dot{x}_2^2 + 2\dot{x}_1 \dot{x}_2) - f_{x1}(\dot{x}_2), -a_3 \sin(x_1) \dot{x}_2^2 - f_{x2}(\dot{x}_1)]$. We find that f_{x1} in (20) includes the states x_1 , $\dot{x}_1$, $\dot{x}_2$, and x_3 . However, the backstepping methods or backstepping-based methods require that f_{x1} includes at most x_1 and $\dot{x}_1$ (Liu et al., 2023a; Rashad et al., 2016; Yu et al., 2018). Moreover, the terms of both $\dot{x}_2$ and x_3 in f_{x1} cannot be separated and considered as a disturbance because it does not satisfy the boundedness assumption (Rashad et al., 2016). Meanwhile, the above also happens for the first-order system of the PDFJ (Wang et al., 2021b), which we will not repeat here. Therefore, conventional backstepping and backstepping-based methods cannot be applied directly to system (20).

Based on (13), we find a direct relationship between the underactuated system dynamics τ_v and the kinetic energy derivative of the two links $\dot{E}_{k2}$. Hence, we can integrate the first two subsystems into one subsystem by considering this relationship in the Lyapunov function of the integrated subsystem. Thereby the number of subsystems is reduced to three, which can alleviate the differential explosion to a large extent in the systematic design. It is worth mentioning that the restriction of the conventional backstepping method (i.e., the relative degree of each virtual subsystem must be one (Kokotovic et al., 1992)) is relaxed.

Furthermore, to tackle the effects of parameter perturbation, model uncertainty, external disturbance, and some remainder terms that are difficult to calculate, we design NN approximators and virtual damping for the controller.

For the three virtual subsystems, we introduce z_1 , z_2 , and z_3 as corresponding errors, respectively:

$$z_1 = [z_{1a}, z_{1b}]^T = [q_1 - q_{1d}, q_2]^T \quad (22)$$

$$z_2 = \theta_1 - u_1 \quad (23)$$

$$z_3 = s_1 \dot{\theta}_1 - u_2 \quad (24)$$

in which $s_1 > 0$ is a scalar. We introduce u_1 , u_2 , and u_3 as the virtual controllers. Based on the basic idea of the backstepping method, the last virtual controller u_3 is equal to the controller τ_1 . The virtual controllers u_1 and u_2 are designed as

$$u_1 = q_{1d} - [c_1 \dot{z}_{1a} + c_1 \beta_1 \tanh(\dot{z}_{1a})] \quad (25)$$

$$u_2 = s_1 c_1 \lambda_0 (Q_A + \hat{Q}_B) - \dot{z}_{1a} - c_2 z_2 \quad (26)$$

where c_1 and c_2 are positive design scalars, β_1 is a design scalar that satisfies $\beta_1 \geq 0$, and

$$\lambda_0 = 1 + \beta_1 \operatorname{sech}^2(\dot{z}_{1a}) \quad (27)$$

$$Q_A = \xi_1 M^{-1}(q_2)[C(q, \dot{q})\dot{q} + P(q, \theta_1)] \quad (28)$$

$$\hat{Q}_B = \hat{K}_1(v_1), \quad v_1 = [q^T, \dot{q}^T, \theta_1]^T \quad (29)$$

We design the following adaptive law with positive design scalars γ_1 and ϖ_1 for the NN weight matrix W_1 of $\hat{Q}_B$:

$$\dot{W}_1 = -\gamma_1 z_2 \lambda_0 G_1(v_1) - \gamma_1 \varpi_1 z_2 W_1 / (K_1 s_1 c_1). \quad (30)$$

The adaptive neural modified-backstepping controller τ_1 is designed as

$$\tau_1 = -(-\hat{\Phi} + I_1 \lambda_1 + I_1 K_1 z_2 + c_3 z_3) / s_1 - s_1 D_a \quad (31)$$

where

$$D_a = \beta_2^2 z_3 / (2\kappa) \quad (32)$$

$$\hat{\Phi} = \hat{K}_2(v_2), \quad v_2 = [q^T, \dot{q}^T, \theta_1, \dot{\theta}_1]^T \quad (33)$$

$$\begin{aligned} \lambda_1 &= c_2 \dot{\theta}_1 + s_1 K_1 \cdot (z_{1a} + q_{1d} - \theta_1) / I_1 - (1 + c_1 c_2 \lambda_0) Q_A \\ &\quad - 2s_1 c_1 \beta_1 \tanh(\dot{z}_{1a}) \operatorname{sech}^2(\dot{z}_{1a}) \cdot (Q_A^2 + Q_A \hat{Q}_B) \end{aligned} \quad (34)$$

$\kappa > 0$, $c_3 > 0$, and $\beta_2 > 0$ are design scalars. We design the following adaptive laws with positive design scalars γ_2 and ϖ_2 for the NN weight matrix W_2 of the unknown function $\hat{\Phi}$:

$$\dot{W}_2 = -\gamma_2 z_3 G_2(v_2) - \gamma_2 \varpi_2 W_2. \quad (35)$$

Besides, the design parameters c_2 and c_3 should be chosen to satisfy the following conditions:

$$c_2 > 1 / (c_1 + \rho_{v1} / K_1) \quad (36)$$

$$c_3 > K_1 / c_A - \{K_1^2 [1 + K_1^2 s_1^2 c_1^2 (1 + \beta_1)^2] / c_A^2 + 1\} / 2 \quad (37)$$

where

$$c_A = c_2 - 1 / (c_1 + \rho_{v1} / K_1). \quad (38)$$

We present the structure diagram of the adaptive neural modified-backstepping control method for the PDFJ in Fig. 2. Clearly, this closed-loop control system has a recursive structure, and it mainly contains two virtual controllers, two NN approximators, and a virtual damping.

3.2. Stability analysis and controller design

This subsection gives a theorem to guarantee the boundedness of $q_1 - q_{1d}$ and q_2 using the adaptive neural modified-backstepping controller and further proves the stability of the whole closed-loop system. Meantime, the design procedure of this controller is included in the stability analysis.

Theorem 1. For the system (5), if the control torque is designed using the adaptive neural modified-backstepping controller (31) under the conditions (36) and (37), and v_j remains within a sufficiently large compact set Ω_j ($j = 1, 2$), then $q_1 - q_{1d}$ is uniformly ultimately bounded, and q_2 converges to zero.

Here, we give the concrete design procedure of the neural modified-backstepping controller and the proof of the system stability.

Proof of Theorem 1. The controller is designed in three steps, and the deduction process of each step is presented as follows.

Step 1. Based on the position control objective, we introduce the error vector z_1 as (22). Considering the kinetic energy of the two links E_{k2} into the Lyapunov function, we choose the Lyapunov candidate function of this first subsystem to be

$$V_1 = \frac{1}{2} z_1^T \begin{bmatrix} \hat{K}_1 & 0 \\ 0 & \hat{K}_2 \end{bmatrix} z_1 + \frac{1}{2} \dot{z}_1^T \hat{M}(q) \dot{z}_1 \quad (39)$$

From Property 1 and (22), it is clear that, except for $V_1(0) = 0$, $V_1 > 0$. According to (22) and the relationship (13) between the underactuated system dynamics τ_v and the kinetic energy derivative of the two links $\dot{E}_{k2}$, the time derivative of V_1 becomes

$$\dot{V}_1 = z_1^T \begin{bmatrix} \hat{K}_1 & 0 \\ 0 & \hat{K}_2 \end{bmatrix} \dot{z}_1 + \dot{z}_1^T \tau_v \quad (40)$$

From (6), (14), (22), and (40), we obtain that

$$\begin{aligned} \dot{V}_1 &= z_1^T \begin{bmatrix} \hat{K}_1 & 0 \\ 0 & \hat{K}_2 \end{bmatrix} \dot{z}_1 - \dot{z}_1^T \begin{bmatrix} \hat{K}_1 \cdot (z_{1a} + q_{1d} - \theta_1) + f_1(\dot{z}_{1a}) \\ \hat{K}_2 z_{1b} + f_2(\dot{z}_{1b}) \end{bmatrix} \\ &= \dot{z}_{1a} [\hat{K}_1 \cdot (\theta_1 - q_{1d}) - f_1(\dot{z}_{1a})] - \dot{z}_{1b} f_2(\dot{z}_{1b}). \end{aligned} \quad (41)$$

We introduce an error variable z_2 that satisfies (23). Based on (41) and the above analysis, we design the virtual controller u_1 of (23) as (25). Substituting (7), (23), and (25) into (41) yields

$$\begin{aligned} \dot{V}_1 &= -\hat{K}_1 [c_1 \dot{z}_{1a}^2 - \dot{z}_{1a} z_2 + c_1 \beta_1 \dot{z}_{1a} \tanh(\dot{z}_{1a})] \\ &\quad - (\rho_{v1} \dot{z}_{1a}^2 + \rho_{c1} |\dot{z}_{1a}| + \rho_{v2} \dot{z}_{1b}^2 + \rho_{c2} |\dot{z}_{1b}|) \end{aligned} \quad (42)$$

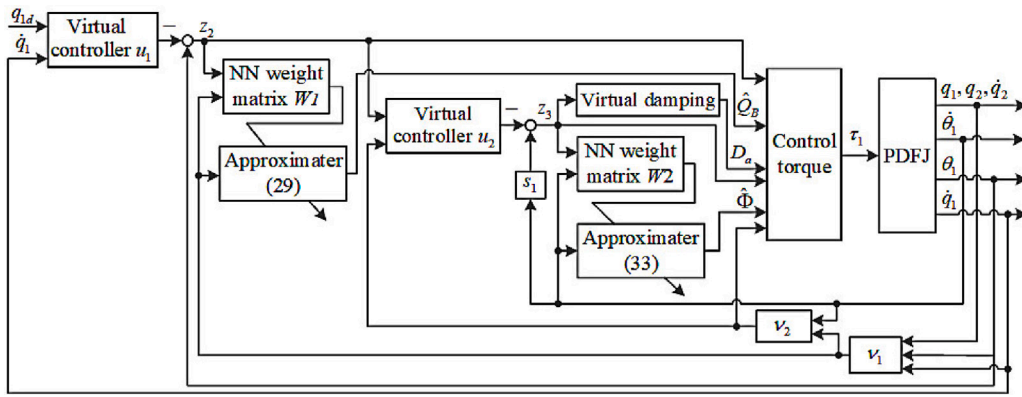


Fig. 2. Structure diagram of the adaptive neural modified-backstepping control method.

Based on (19) and (42), $\dot{V}_1$ satisfies that

$$\begin{aligned}
\dot{\gamma}_1 &\leq -\dot{K}_1[c_1\dot{z}_{1a}^2 - |\dot{z}_{1a}||z_2| + c_1\beta_1|\dot{z}_{1a}\tanh(\dot{z}_{1a})|] \\
&\quad -(\rho_{v1}\dot{z}_{1a}^2 + \rho_{c1}|\dot{z}_{1a}| + \rho_{v2}\dot{z}_{1b}^2 + \rho_{c2}|\dot{z}_{1b}|) \\
&\leq -\dot{K}_1|\dot{z}_{1a}|[(c_1 + \rho_{v1}/\dot{K}_1)|\dot{z}_{1a}| - (|z_2| - \beta_\lambda)] - \rho_{v2}\dot{z}_{1b}^2 - \rho_{c2}|\dot{z}_{1b}| \\
&\leq -\dot{K}_1|\dot{z}_{1a}|[(c_1 + \rho_{v1}/\dot{K}_1)(|\dot{z}_{1a}| - \beta_{V1})] - \rho_{v2}\dot{z}_{1b}^2 - \rho_{c2}|\dot{z}_{1b}| \\
\beta_\lambda &= \rho_{c1}/\dot{K}_1 + c_1\beta_1|\tanh(\dot{z}_{1a})| \\
\beta_{V1} &= (|z_2| - \beta_\lambda)/(c_1 + \rho_{v1}/\dot{K}_1)
\end{aligned} \tag{43}$$

From (43), when $|z_{1a}| \geq \beta_{V1}$, $\dot{V}_1 \leq 0$, and then $|z_{1a}|$ will decrease to less than β_{V1} or remain the same. When $|z_{1a}| < \beta_{V1}$, $\dot{V}_1 > 0$, $|z_{1a}|$ will increase until $|z_{1a}| \geq \beta_{V1}$ again. Therefore, (43) implies that there exists an unknown time constant t_1 ($t_1 > 0$) such that when $t \geq t_1$, z_{1a} is bounded by

$$|\dot{z}_{1a}| \leq (|z_2| - \beta_A) / [c_1 + \rho_{v1} / \hat{K}_1] \quad (44)$$

Besides, we further obtain that $|z_2| \geq \beta_A$ because the denominator in (43) is positive and $|z_{1a}| \geq 0$. Through construction of the Lyapunov candidate function (39) and the analysis based on the Lyapunov method (Khalil, 2002), we obtain that z_{1a} is limited to a boundary related to the variable z_γ .

Step 2. Taking the time derivative of z_2 yields

$$\dot{z}_2 = \dot{\theta}_1 + c_1 \lambda_0 \ddot{z}_{1a} \quad (45)$$

However, in practice, the acceleration information $\ddot{z}_{1a}$ is usually unmeasurable. Combining (22) and (5b), we write $\ddot{z}_{1a}$ to be

$$\ddot{z}_{1q} = -[Q_A(q, \dot{q}, \theta_1) + Q_B(q, \dot{q}, \theta_1)] \quad (46)$$

where $Q_A(q, \dot{q}, \theta_1)$ is showed in (28), and

$$\mathcal{Q}_B(q, \dot{q}, \theta_1) = \xi_1 \hat{M}^{-1}(q_2)[\hat{C}(q, \dot{q})\dot{q} + \hat{P}(q, \theta_1) + \Psi_q(\dot{q})] - \mathcal{Q}_A(q, \dot{q}, \theta_1) \quad (47)$$

For the sake of convenience in writing, we abbreviate $Q_A(q, \dot{q}, \theta_1)$ and $Q_B(q, \dot{q}, \theta_1)$ to Q_A and Q_B , respectively. Because $\Psi_q(\dot{q})$ in Q_B is unknown, we utilize a NN for Q_B to calculate its approximation term $\hat{Q}_B$ as (29).

Next, we choose the second candidate Lyapunov function V_2 to be

$$V_2 = s_1 K_1 z_2^2 / 2 + K_1 s_1 c_1 \tilde{W}_1^T \tilde{W}_1 / (2\gamma_1) \quad (48)$$

Based on (18) and (45)–(48), the time derivative of V_2 becomes

$$\dot{V}_2 = K_1 s_1 \{-c_1 \tilde{W}_1^T \dot{W}_1 / \gamma_1 + z_2 [\dot{\theta}_1 - c_1 \lambda_0 \cdot (Q_A + Q_B)]\} \quad (49)$$

We introduce an error variable z_3 that satisfies (24). Based on (41) and the above analysis, we design the virtual controller u_2 of (24) as (26).

Substituting (24) and (26) into (49) yields

$$\begin{aligned} \dot{V}_2 = & -K_1(c_2 z_2^2 + \dot{z}_{1a} z_2 - z_2 z_3 + s_1 c_1 z_2 \lambda_0 \varepsilon_1) \\ & -K_1 s_1 c_1 \tilde{W}_1^T [z_2 \lambda_0 G_1(v_1) + \dot{W}_1 / \gamma_1] \end{aligned} \quad (50)$$

Based on (50), we design the adaptive law of the NN weight matrix W_1 as (30). From (19), (30), (43), (44), (50), and Assumption 2, when

$t \geq t_1$, we obtain

$$\begin{aligned} \dot{V}_2 &= -K_1(c_2 z_2^2 + \dot{z}_{1a} z_2 - z_2 z_3 + s_1 c_1 z_2 \lambda_0 \varepsilon_1) + \varpi_1 z_2 \tilde{W}_1^T W_1 \\ &\leq K_1 |z_2| [-c_2 |z_2| + (|\dot{z}_{1a}| + |z_3| + \lambda_2)] \\ &\leq K_1 |z_2| [-c_A |z_2| + (-c_B + |z_3| + \lambda_2)] \end{aligned} \quad (51)$$

where c_A is written as (38),

$$\begin{aligned} \lambda_2 &= K_1 s_1 c_1 [1 + \beta_1 \operatorname{sech}^2(\dot{z}_{1a})] |\epsilon_{1M}| + \varpi_1 \|W_1^*\|^2/2 - \varpi_1 \|\tilde{W}_1\|^2/2 \\ c_B &= (\rho_{c1}/\hat{K}_1 + c_1 \beta_1 |\tanh(\dot{z}_{1a})|)/(c_1 + \rho_{v1}/\hat{K}_1) \end{aligned} \quad (52)$$

We choose c_2 to satisfy the condition (36) so as to make $c_4 > 0$. Hence, there exists an unknown time constant t_2 ($t_2 > 0$) such that after a sufficiently long time $t \geq t_2$, it is calculated from (51) that

$$|z_2| \leq (-c_B + |z_3| + \lambda_2)/c_A \quad (53)$$

Based on (44) and (51), we find that z_2 converges only after $t \geq t_2 \geq t_1$.

Step 3. Based on (5a), (6), (19), (26), (45), and (46), taking the time derivative of z_3 yields

$$\dot{z}_3 = \lambda_1 + s_1(\tau_1 - f_0 - \sigma_0(t))/\hat{I}_1 - \Phi_0 \quad (54)$$

where λ_1 equals (34), f_0 is the abbreviation of $f_0(q_1, \theta_1, \dot{\theta}_1)$,

$$\begin{aligned} \Phi_0 = & 2s_1c_1\beta_1 \tanh(\hat{z}_{1a})\text{sech}^2(\hat{z}_{1a})(Q_AQ_B + Q_B\hat{Q}_B) \\ & + (1 + c_2c_1\lambda_0)Q_B + s_1c_1\lambda_0(\dot{Q}_A + \dot{\hat{Q}}_B) - s_1(\hat{K}_1/\hat{I}_1 \\ & - K_1/I_1)(z_{1a} + q_{1d} - \theta_1). \end{aligned} \quad (55)$$

Then, we choose the third Lyapunov candidate function V_3 as

$$V_3 = \frac{1}{2} \hat{I}_1 z_3^2 + \frac{1}{2\gamma_2} \tilde{W}_2^T \tilde{W}_2 \quad (56)$$

Combining (18) and (54), the time derivative of V_3 satisfies

$$\begin{aligned}\dot{V}_3 &= z_3[\lambda_1 I_1(1+k_{I_1}) + s_1(\tau_1 - f_0 - \sigma_0(t)) - \hat{I}_1 \Phi_0] - \tilde{W}_2^T \dot{W}_2 / \gamma_2 \\ &= z_3[\lambda_1 I_1 + s_1(\tau - \sigma_0(t)) - \Phi] - \tilde{W}_2^T \dot{W}_2 / \gamma_2 \\ \Phi &= -k_{I_1} I_1 \lambda_1 + s_1 f_0 + \hat{I}_1 \Phi_0\end{aligned}\quad (57)$$

Finally, from (57), we can design an adaptive neural modified-backstepping controller τ_1 as (31), where D_a in (32) is the virtual damping term utilized to overcome $\sigma_0(t)$, and $\hat{\Phi}$ in (33) is the approximation of the unknown function Φ .

By substituting the controller (31) into (57), we obtain that

$$\dot{V}_3 = -I_1 K_1 z_2 z_3 - c_3 z_3^2 + \lambda_3 + s_1 \lambda_4 \quad (58)$$

where

$$\begin{aligned}\lambda_3 &= -z_3(\Phi - \hat{\Phi}) - \tilde{W}_2^T \dot{W}_2 / \gamma_2 \\ \lambda_4 &= -\sigma_0(t)z_3 - \beta_3^2 z_3^2 / (2\kappa)\end{aligned}\quad (59)$$

From (15), (17), (18), and (33), λ_3 becomes

$$\lambda_3 = -z_3 \varepsilon_2 - \tilde{W}_2^T [\dot{W}_2 / \gamma_2 + z_3 G_2(v_2)] \quad (60)$$

Based on (60), we design the adaptive law of the NN weight matrix W_2 as (35). Based on (35), (60), and Assumption 2, λ_3 satisfies that

$$\lambda_3 = -z_3 \varepsilon_2 + \varpi_2 \tilde{W}_2^T W_2 \leq |z_3| |\varepsilon_{2M}| + \varpi_2 (\|\tilde{W}_2^*\|^2 - \|\tilde{W}_2\|^2)/2 \quad (61)$$

Based on Assumption 1, (32), and (59), there must exist an auxiliary positive constant k_β such that $\beta_2 k_\beta \geq \sigma_m \geq |\sigma_0|$. Therefore, λ_4 satisfies

$$\begin{aligned} \lambda_4 &= -\sigma_0(t) z_3 - (\beta_2 k_\beta)^2 z_3 / (2\kappa k_\beta^2) \\ &\leq |\beta_2 k_\beta| \cdot |z_3| - [\beta_2^2 k_\beta^2 z_3^2 / (2\kappa k_\beta^2) + 2\kappa k_\beta^2 / 4] + \kappa k_\beta^2 / 2 \\ &\leq |\beta_2 k_\beta| \cdot |z_3| - |\beta_2 k_\beta z_3| + \kappa k_\beta^2 / 2 \\ &\leq \kappa k_\beta^2 / 2 \end{aligned} \quad (62)$$

Then, based on (53), (58), (61)–(62), after a sufficiently long time that $t > t_1$, we obtain that

$$\dot{V}_3 \leq -Y_1 z_3^2 - Y_2 \|\tilde{W}_2\|^2 + \Delta \quad (63)$$

where

$$\begin{aligned} Y_1 &= c_3 - I_1 K_1 / c_A - \{K_1^2 [1 + K_1^2 s_1^2 c_1^2 (1 + \beta_1)^2] / c_A^2 + 1\} / 2 \\ Y_2 &= \varpi_2 / 2 \\ \Delta &= (I_1^2 |\varepsilon_{1M}|^2 + I_1^2 \varpi_1^2 \|\tilde{W}_1^*\|^4 / 4 + |\varepsilon_{2M}|^2 + \varpi_2 \|\tilde{W}_2^*\|^2 \\ &\quad + s_1 \kappa k_\beta^2) / 2 - \varpi_1 K_1 |z_3| \|\tilde{W}_1\|^2 / 2 - c_B K_1 |z_3| \\ &\leq (I_1^2 |\varepsilon_{1M}|^2 + I_1^2 \varpi_1^2 \|\tilde{W}_1^*\|^4 / 4 + |\varepsilon_{2M}|^2 + \varpi_2 \|\tilde{W}_2^*\|^2 + s_1 \kappa k_\beta^2) / 2 \end{aligned} \quad (64)$$

From (64), the terms about $|z_3|$ in Δ are non-positive and play a role in narrowing Δ , and then Δ is limited by a constant boundary. We choose c_3 to satisfy the condition (37), and then obtain that $Y_1 > 0$. Meantime, it is clear that $Y_2 > 0$.

According to (56), V_3 can be rewritten to be

$$V_3 = Y_A \Lambda, \quad Y_A = [1/(2I_1), 1/(2Y_2)], \quad \Lambda = [z_3^2, \tilde{W}_2^T \tilde{W}_2]^T \quad (65)$$

Let Y_M be the minimal element in Y_A . It can be seen that $Y_M \|\Lambda\| \leq V_3$. Next, let $Y_B = [Y_1, Y_2]$. Based on (63) and (65), there exist positive scalars Y_N and Δ_N such that

$$\dot{V}_3 \leq -Y_B \Lambda + \Delta \leq -Y_N V_3 + \Delta_N. \quad (66)$$

Lemma 2 (Ge and Wang (2004)). Let ζ be a column vector of states, and let $V(\zeta)$ be the positive and continuously differentiable Lyapunov function, wherein $V(\zeta(0))$ is bounded. There exists a positive scalar α_1 such that $\alpha_1 \|\zeta\|^2 \leq V(\zeta)$. If the inequality $\dot{V}(\zeta) \leq -\alpha_2 V(\zeta) + \alpha_3$ with $\alpha_2 > 0$ and $\alpha_3 > 0$ holds, ζ will be uniformly ultimately bounded, and the solution of ζ will exponentially converge to compact set $\Omega_\zeta = \{\zeta | \|\zeta\| \leq \sqrt{\alpha_3 / (\alpha_1 \alpha_2)}\}$.

Based on (65), (66), and Lemma 2, we know that z_3 and $\tilde{W}_2$ in the closed-loop system are ultimately uniformly bounded and

$$\lim_{t \rightarrow \infty} |z_3| \leq \beta_C, \quad \beta_C = \sqrt{\Delta_N / (Y_M Y_N)} \quad (67)$$

From (53) and (67), z_2 is ultimately uniformly bounded, and

$$\lim_{t \rightarrow \infty} |z_2| \leq \beta_B, \quad \beta_B = (-c_B + \beta_C + \lambda_2) / c_A \quad (68)$$

From (43) and (68), we obtain that

$$\beta_B - \beta_A = [\beta_C + \lambda_2 - c_2 \rho_{c1} / \hat{K}_1 - c_1 c_2 \beta_1 |\tanh(\dot{z}_{1a})|] / c_A \quad (69)$$

Based on (38), (52), (64), and Assumptions 1 and 2, we find out that $c_A > 0$, and β_C and λ_2 are bounded. Therefore, from (69), we can select large enough c_1 , c_2 , and β_1 to make $\beta_B < \beta_A$ under the conditions (36)–(38). Therefore, it is further derived from (43) that

$$\lim_{t \rightarrow \infty} \dot{V}_1 \leq 0. \quad (70)$$

We define Ω_f to be the largest invariant set such that $\dot{V}_1 = 0$. Based on (43), when $\dot{V}_1 \equiv 0$, it can be deduced that $\dot{z}_{1a} \equiv 0$ and $\dot{z}_{1b} \equiv 0$. At this time, we define $q_1 = q_1^*$, $q_2 = q_2^*$, $z_{1a} = z_{1a}^*$, and $z_2 = z_2^*$. So, $\dot{q}_1^* \equiv 0$, $\dot{q}_2^* \equiv 0$, $\dot{q}_1^* \equiv 0$, $\dot{q}_2^* \equiv 0$, and q_1^* and q_2^* are constants. Applying the above results and substituting (22) and (25) into (5b) yield

$$\begin{bmatrix} \ddot{q}_1^* \\ \ddot{q}_2^* \end{bmatrix} = \hat{M}^{-1}(q_2^*) \begin{bmatrix} -\hat{K}_1 z_{1a}^* + \hat{K}_1 z_2^* \\ -\hat{K}_2 q_2^* \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (71)$$

When (71) holds for all time,

$$q_1^* - q_{1d} = z_{1a}^* = z_2^*, \quad q_2^* = 0 \quad (72)$$

Thus, based on (68) and LaSalle's invariance principle, the following conclusions are obtained:

$$\begin{aligned} \lim_{t \rightarrow \infty} (q_1 - q_{1d}) &= z_2 \leq \beta_B, \quad \lim_{t \rightarrow \infty} q_2 = 0 \\ \lim_{t \rightarrow \infty} \dot{q}_1 &= \dot{z}_{1a} = 0, \quad \lim_{t \rightarrow \infty} \dot{q}_2 = 0 \end{aligned} \quad (73)$$

and $q_1 - q_{1d}$ is ultimately uniformly bounded. ■

Therefore, $q_1 - q_{1d}$ converges to a small neighborhood of zero. From this conclusion and (73), the position control objective of the PDFJ is achieved.

Remark 1. Based on (43) and (68), when $\rho_{c1} / \hat{K}_1 > \beta_B$ is satisfied, the results in (70) are also obtained, and then the control objective is achieved. At this time, if we choose $\beta_1 = 0$, the controller τ_1 can be further simplified.

Remark 2. We provide the selection principles for design parameters in the proposed controller (31) by the following four steps, and the design parameters include c_1 , c_2 , c_3 , s_1 , β_1 , β_2 , γ_1 , γ_2 , ϖ_1 , ϖ_2 , κ , ϑ_j , b_{1r1} , b_{2r2} , η_1 , and η_2 .

Step 1. c_1 , c_2 , and c_3 need to be chosen under conditions (36) and (37).

Step 2. We give some ways to improve the control accuracy of the first link through the selection of design parameters. Based on (32), (38), (48), (52), (64), (65), (67), (68), and (73), the boundary of $|q_1 - q_{1d}|$ is reduced by making $\gamma_1 \geq 1$, $\gamma_2 \geq 1$, and $s_1 \geq 1$ and choosing a larger c_2 and the smaller c_1 , s_1 , β_1 , κ , $1/\beta_2$, ϖ_1 , and ϖ_2 .

Step 3. From (24), (26), (31), (38), (63), (67), (68), and (73), we select large ϖ_2 , c_2 , and c_3 as well as small s_1 , c_1 , and β_1 , thus enabling the PDFJ to reach the target with short vibration and fast convergence speed. Note that these design parameters need to be selected within a certain range, otherwise the system is prone to angular overshoot and an initial peak in torque. Meanwhile, from the controller (31), we set $c_3 = c_{30} / I_1$ and avoid excessively small κ and $1/\beta_2$, so as to further avoid the phenomenon of torque peaking.

Step 4. The Ge et al. (2013) provides a very detailed study on the selection of design parameters (i.e., ϑ_j , b_{1r1} , b_{2r2} , η_1 , and η_2) for RBF neural network, we will not repeat it here.

It is clear that the presented controller (31) takes into account the states of two links. By selecting the design parameters according to the above principles, the vibration of the first link is quickly suppressed. Meanwhile, since the second link is only connected to the passive joint at the end of the first link, its vibration is subsequently quickly suppressed, and then q_2 converges to 0 by the effect of friction.

4. Simulations and experiments

In this section, we present the simulation and experimental results to verify the validity of the adaptive neural modified-backstepping method. To show the advantages of the proposed method, we compare it with the existing methods.

4.1. Simulation results

In this section, we use the Matlab/Simulink software to verify the validity of the proposed method. The nominal physical parameters of the PDFJ are given as $m_1 = 0.3$ kg, $m_2 = 0.15$ kg, $L_1 = 0.3$ m, $L_2 = 0.25$ m, $J_1 = 0.02$ kg · m², $J_2 = 0.01$ kg · m², $K_1 = K_2 = 0.8$ N · m/rad, $I_1 = L_1/2$, $I_2 = L_2/2$, and $I_1 = 5 \times 10^{-5}$ kg · m². We assume that the actual physical parameters fluctuate within 30% of the nominal physical parameters. Here, the perturbed percentages of the physical parameters are set as $k_{I_1} = k_{m_1} = k_{K_1} = k_{J_1} = k_{L_1} = k_{L_2} = 30\%$.

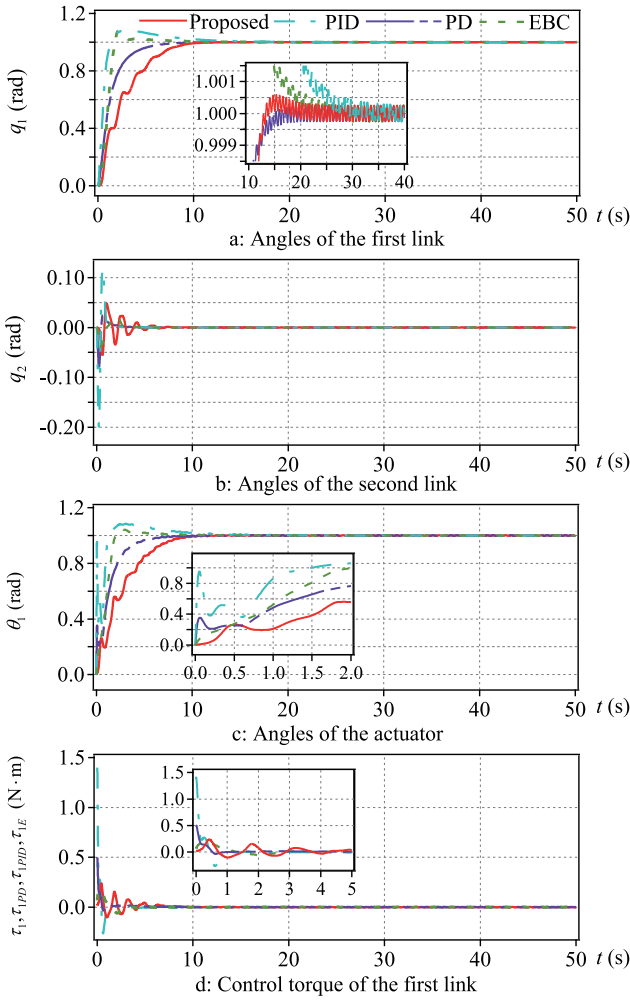


Fig. 3. The first simulation results on position control of the PDFJ using the proposed method, the EBC method, the PD method, and the PID method.

In the simulation case, we set the unknown friction coefficients, unknown model uncertainty, and external disturbance of the model (5) to be $\rho_{vi} = 0.02$, $\rho_{ci} = 0.01$,

$$\begin{aligned} f_0(q_1, \theta_1, \dot{\theta}_1) &= 0.02\dot{\theta}_1 + 0.01\tanh(\rho_h\dot{\theta}_1) + 0.01\cos(q_1)\dot{\theta}_1 \\ \sigma_0(t) &= 0.005\sin(10t) \end{aligned} \quad (74)$$

in which $\rho_h = 20$. To alleviate the effect of the signum function from the dynamic model (5) on Matlab computing speed, we employ the hyperbolic tangent function $\tanh(\rho_h*)$ instead of the signum functions $\text{sgn}(*)$ in (5).

So far, there have been few studies on the control of the PDFJ. Since the energy-based control (EBC) method (Lai et al., 2000; Liu & Xin, 2017), the conventional proportion-derivative (PD) method, and the proportion-integral-derivative (PID) method are well-known methods and are capable of controlling some underactuated non-strict-feedback systems with uncertainties, we compare our proposed method with them to demonstrate the superiority of our presented method. Based on the following Lyapunov function candidate that

$$V_e = \frac{k_{e1}}{2}(\theta_1 - q_{1d})^2 + \frac{1}{2}(\dot{\theta}_1 - \dot{q}_{1d})^2 + E_{k2} + E_p \quad (75)$$

the EBC controller is designed as

$$\tau_{1E} = -k_{e1}(\theta_1 - q_{1d}) - \frac{K_1(q_1 - \theta_1) + k_{e2}\tanh(k_{e3}\dot{\theta}_1)}{1 + I_1} \quad (76)$$

where k_{e1} , k_{e2} , and k_{e3} are positive constants. The PID controller and the PD controller are designed as

$$\begin{aligned} \tau_{1PD} &= -k_{p1}\dot{z}_{1a} - k_{d1}\dot{z}_{1a} \\ \tau_{1PID} &= -k_{p2}z_{1a} - k_{i2}\int_0^t z_{1a}dt - k_{d2}\dot{z}_{1a}. \end{aligned} \quad (77)$$

For a fair comparison, we restrict the design parameters of the proposed method, the EBC method, the PD method, and the PID method, selected after several tests, to satisfy the following strict but reasonable principles: (1) The overshoot of the first link angle q_1 is less than 10% of q_{1d} ; (2) For all $t \geq 25$ s, the positioning accuracy of the first link satisfies that $|q_1 - q_{1d}| \leq 0.001$ rad.

Based on Remark 2, the above principles, and plenty of simulations, we choose the design parameters of the four control methods as follows. For the proposed method, the design parameters of the RBF NN approximators are chosen to be $\theta_j = 5$, $b_{1r1} = 0.4$, $b_{2r2} = 1$, $w_{jrj}(0) = 0$, $\eta_1 = [q_{1d}, 0, 0, 0, q_{1d}, 0]^T \mu_1 + [0, i_1, i_2, i_3, 0, i_4]^T \mu_2$, and $\eta_2 = [q_{1d}, 0, 0, 0, q_{1d}]^T \mu_1 + [0, i_1, i_2, i_3, 0]^T \mu_2$, where $i_1 = 0.1$, $i_2 = 0.005$, $i_3 = 0.015$, $i_4 = 0.1$, $\mu_1 = [0.7, 0.9, 1, 1.1, 1.3]$, and $\mu_2 = [-5, -1, 0, 1, 5]$ ($r_j = 1, 2$; $j = 1, 2$; $n = 2$). The parameters of the controller (31) and the parameters of the adaptive laws (30) and (35) are set to be $c_1 = 2$, $c_2 = 10$, $c_3 = 10I_1$, $s_1 = 1$, $\beta_1 = 0.3$, $\beta_2 = 0.045$, $\gamma_1 = 1$, $\gamma_2 = 1$, $\varpi_1 = 3$, $\varpi_2 = 5$, and $\kappa = 0.5$. The design parameters of the EBC method are set to be $k_{e1} = 0.07$, $k_{e2} = 0.001$, and $k_{e3} = 20$. For the PD method and the PID method, the design parameters are chosen as $k_{p1} = 0.6$, $k_{d1} = 0.7$, $k_{p2} = 1.4$, $k_{d2} = 0.8$, and $k_{i2} = 0.27$.

We quantify the control performance of the proposed method, the PD method, and the PID method with the following five performance indices:

- (1) the settling time t_s (i.e., $|q_1 - q_{1d}| \leq 0.001$ rad, $\forall t \geq t_s$);
- (2) the overshoot $\hat{h}_o$ of q_1 (i.e., $\hat{h}_o = |(q_1 - q_{1d})/q_{1d}|$);
- (3) the steady-state error e_{1s} of $|q_1 - q_{1d}|$;
- (4) the steady-state error e_{2s} of $|q_2|$;
- (5) the maximum value τ_{1m} of $|\tau_1|$.

Among these performance indices, the settling time t_s is used to assess the speed at which q_1 achieves $|q_1 - q_{1d}| \leq 0.001$ rad mentioned in the principle (2). Clearly, a good control performance for the PDFJ is usually accompanied by a short t_s and small $\hat{h}_o$, e_{1s} , e_{2s} , and τ_{1m} .

The simulation results and quantified results of the control performance are presented in Fig. 3 and Table 1, respectively. We can find that the adaptive neural modified-backstepping method can make the PDFJ move to the target position with high precision in a suitable torque range and short settling time. This method overcomes the effect of external disturbances and then effectively suppresses the vibration of the two links. Although the proposed method has a longer settling time than the EBC method, the PD method, and the PID method, we find that the EBC method and the PID method have larger overshoot, resulting in longer convergence time, and the PD method has a larger steady-state error. As we are more concerned with designing a controller to make the system have as small a steady-state error and as little overshoot as possible, the proposed method is superior to the other three methods.

Moreover, we present the weight updating process and the approximation results of the RBF NNs in Fig. 4. Note that Q_B and Φ are unknown functions, including some uncertainty terms of the dynamic model. Considering that the unknown uncertainty terms are artificially given in the simulation, to illustrate the approximation capability of the RBF NNs, we forcibly use these terms to calculate Q_B and Φ . As shown in Fig. 4, the outputs of RBF NNs gradually approach the values of Q_B and Φ , which indicates that the RBF NNs that we used have the capacity to approximate the unknown functions. Since $\hat{Q}_B$ multiplies with some design parameters in the controller (31) to co-affect the system performance, there is a trade-off between $\hat{Q}_B$ and these design parameters. Following Step 3 of Remark 2, we select design parameters to improve convergence speed and reduced the trade-off, so as to weaken the effect of the NN estimated error in the initial stage. Although there are some small and bounded approximation errors, the impact of approximation errors on the control performance can be effectively attenuated by c_B

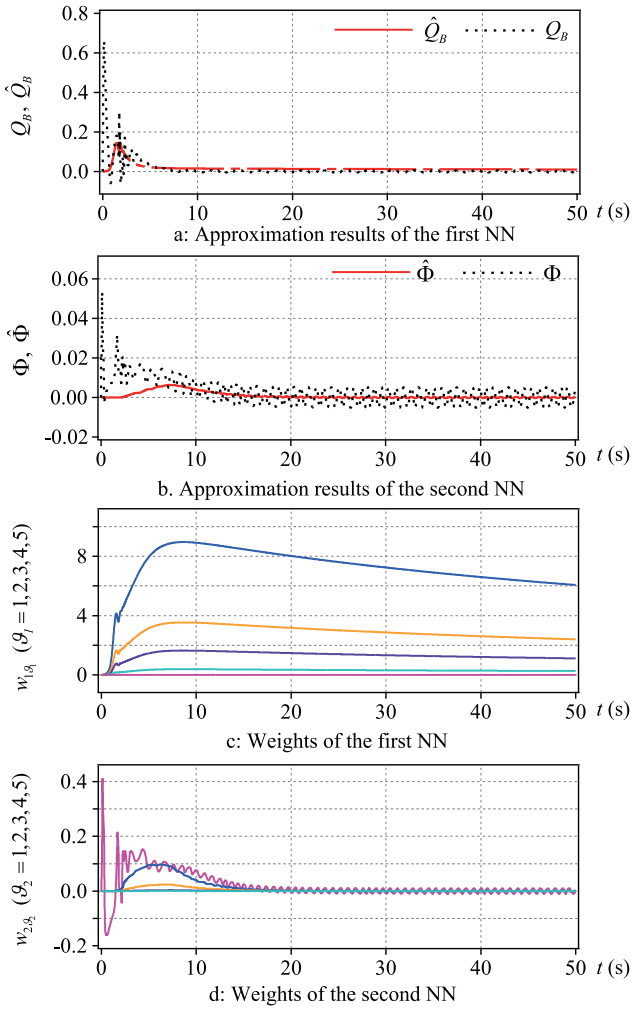


Fig. 4. The approximation results and weights of the RBF NNs.

Table 1
Quantified first simulation results of the control performance.

Methods	t_s (s)	$\hat{h}_o$ (%)	e_{1s} (rad)	e_{2s} (rad)	τ_{1m} (N m)
Proposed	12.1	0.06	2.60×10^{-4}	2.86×10^{-4}	0.24
PD	12.5	0.03	2.67×10^{-4}	3.36×10^{-4}	0.50
PID	23.4	8.30	2.58×10^{-4}	2.99×10^{-4}	1.40
EBC	17.3	8.66	2.93×10^{-4}	2.90×10^{-4}	0.17

and D_a in the proposed method. Furthermore, Fig. 4 shows that the weights of the NNs converge to fixed values, and they are bounded.

To further demonstrate the superiority and robustness of our proposed method, we perform a simulation for the PDFJ with more uncertainty terms, i.e.,

$$\begin{aligned}
 \sigma_0(t) &= 0.01 \sin(10t) + 0.01 \sin(20t + \pi/3) \\
 f_0(q_1, \theta_1, \dot{\theta}_1) &= 0.025\dot{\theta}_1 + 0.02\tanh(30\dot{\theta}_1) \\
 &\quad + 0.015 \cos(q_1)\dot{\theta}_1 - 0.01(\dot{\theta}_1 - \dot{q}_1) \\
 f_1 &= 0.025\dot{q}_1 + 0.02\tanh(30\dot{q}_1) + 0.01(\dot{\theta}_1 - \dot{q}_1) \\
 f_2 &= 0.025\dot{q}_2 + 0.02\tanh(30\dot{q}_2)
 \end{aligned} \quad (78)$$

Meanwhile, we adjust some control parameters as follows so that the proposed and comparative methods still meet the requirements of principles (1) and (2): $c_1 = 1.2$, $c_2 = 25$, $c_3 = 5I_1$, $\beta_1 = 0.85$, $\beta_2 = 0.05$, $k_{p1} = 0.9$, $k_{d1} = 1.8$, $k_{p2} = 1.5$, $k_{i2} = 0.28$, $k_{d2} = 0.75$, $k_{e1} = 0.185$, $k_{e2} = 0.023$, and $k_{e3} = 30$. As shown in Fig. 5 and Table 2, the proposed method makes the PDFJ have better control performance than the PD

Table 2
Quantified second simulation results of the control performance.

Methods	t_s (s)	$\hat{h}_o$ (%)	e_{1s} (rad)	e_{2s} (rad)	τ_{1m} (N m)
Proposed	18.2	0.03	2.73×10^{-4}	4.80×10^{-5}	0.19
PD	19.4	0.06	6.65×10^{-4}	1.17×10^{-4}	0.93
PID	21.6	7.40	5.52×10^{-4}	6.64×10^{-5}	1.53
EBC	24.5	8.96	7.12×10^{-4}	5.08×10^{-5}	0.33

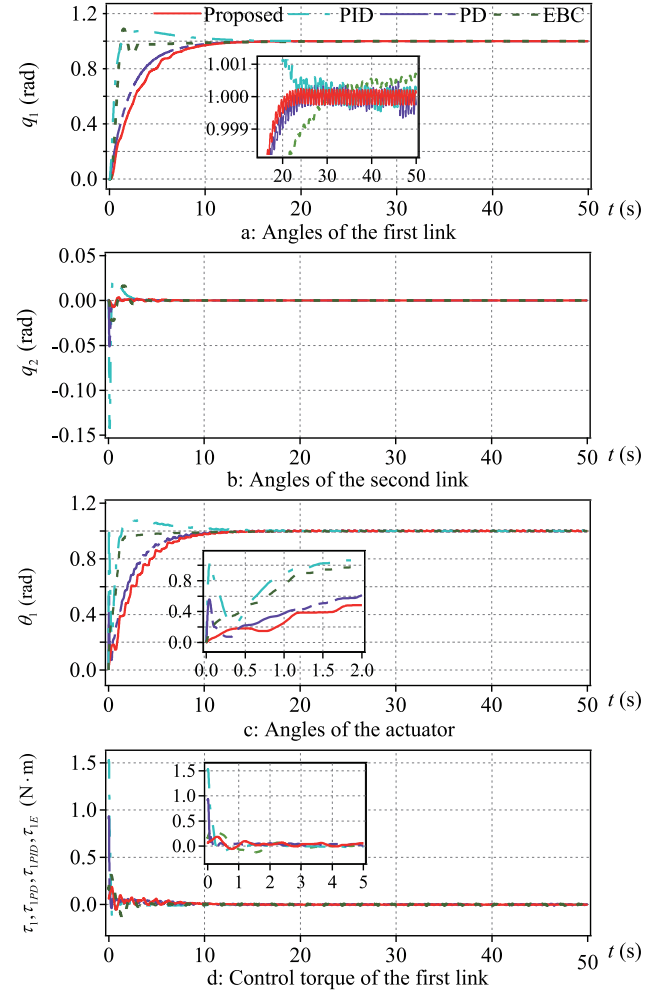


Fig. 5. The second simulation results on position control of the PDFJ using the proposed method, the EBC method, the PD method, and the PID method.

method, the PID method, and the EBC method. Thus, the proposed method is superior to the three methods in controlling the PDFJ.

4.2. Experimental results

Here, we employ an experiment platform of a PDFJ to further verify the effectiveness of the proposed method. As shown in Fig. 6, the flexible joints are made of torsional springs. We present the experimental process as follows. The motor angle and two link angles are measured by a motor incremental encoder (with 2500 pulses per revolution (PPR)) and two incremental photoelectric encoders (with 2500 PPR), respectively. We use an I/O module (National Instrument PCIe-6361) to transmit the measurement signals to the MATLAB/Simulink software in a PC (Intel Core i7-12700K; 32G), and we use the toolbox functions “Encoder Input” in Simulink to attain real-time θ_1 , q_1 , and q_2 . Since all the counters from the encoders work in the quadruple mode, we obtain 10000 PPR counter data, and each pulse number corresponds

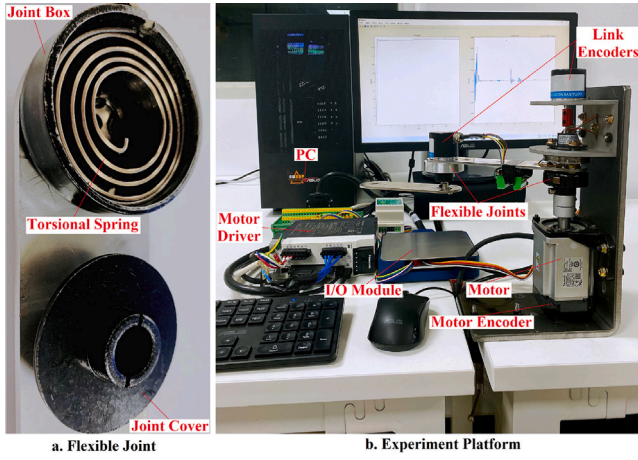


Fig. 6. Experiment platform of the PDFJ.

to 6.283×10^{-4} rad. We attain the derivatives of θ_1 , q_1 , and q_2 by differentiating in Simulink. The program for control torque τ_1 (31) is written into Simulink, in which we set q_1 , q_2 , θ_1 , and their velocities to start from zero, make $q_{1d} = 1$ rad, and set the physical parameters of the PDFJ to be $m_1 = 0.52$ kg, $m_2 = 0.15$ kg, $L_1 = 0.18$ m, $L_2 = 0.16$ m, $I_1 = L_1/2$, $I_2 = L_2/2$, $J_1 = 0.007$ kg $\cdot$ m², $J_2 = 8.2 \times 10^{-5}$ kg $\cdot$ m², $I_1 = 5.6 \times 10^{-5}$ kg $\cdot$ m², and $K_1 = K_2 = 0.15$ N $\cdot$ m/rad. The sampling period of this experiment is set to be 0.001s. Then, we utilize the toolbox function “Analog Output” in Simulink and the I/O module to transfer τ_1 to the Panasonic motor driver MBDLT25SF in the form of a proportional analog voltage signal. On this basis, the motor driver, configured to work in its built-in Torque Control Mode, employs its built-in algorithm to transfer a driver signal to a servo motor (Panasonic MINAS A6 series MHMF042L1U2M) that matches the motor driver, making the motor generate an actual torque approximately equal to τ_1 . This servo motor has a rated torque of 1.27 N $\cdot$ m and an instantaneous maximum torque of 4.46 N $\cdot$ m. With the above settings, the motor drives the flexible joint to rotate and then indirectly drives the two links based on the control torque τ_1 .

The actual PDFJ used in the experiment has large friction and low-stiffness flexible joints. Thus, it requires a large torque to make the PDFJ just start to move. But once the PDFJ moves, its links are easy to vibrate. It is challenging work to control such a manipulator, especially for the PD method. Here, we choose the design parameters of the proposed method and the PID method under the following two principles: (1) The overshoot of the first link angle q_1 is less than 25% of q_{1d} ; (2) For all $t \geq 15$ s, $|q_1 - q_{1d}| \leq 0.02$ rad.

Under the two principles and plenty of experiments, the design parameters of the proposed method are set to be $\theta_j = 5$, $b_{1r_1} = 2$, $b_{2r_2} = 2$, $w_{j_r}(0) = 0$, $\eta_1 = [q_{1d}, 0, 0, 0, q_{1d}, 0]^T \mu_1 + [0, t_1, t_2, t_3, 0, t_4]^T \mu_2$, $\eta_2 = [q_{1d}, 0, 0, 0, q_{1d}]^T \mu_1 + [0, t_1, t_2, t_3, 0]^T \mu_2$, where $t_1 = 0.1$, $t_2 = 0.3$, $t_3 = 0.05$, $t_4 = 0.5$, $\mu_1 = [-0.5, 0, 0.5, 1, 1.5]$, and $\mu_2 = [-5, -2, 0, 2, 5]$ ($r_j = 1, 2$; $j = 1, 2$; $n = 2$). The parameters of the controller (31) and the parameters of the adaptive laws (30) and (35) are chosen as $c_1 = 1.5$, $c_2 = 4$, $c_3 = 4I_1$, $s_1 = 1$, $\beta_1 = 0.2$, $\beta_2 = 0.02$, $\gamma_1 = 0.1$, $\gamma_2 = 1$, $\varpi_1 = 1$, $\varpi_2 = 1$, and $\kappa = 0.6$. Besides, we choose the design parameters of the PID method as $k_{p2} = 0.4$, $k_{d2} = 0.2$, and $k_{j2} = 0.6$. As for the PD method, we select a set of best control parameters that $k_{p1} = 0.6$ and $k_{d1} = 0.25$.

The experimental results are shown in Figs. 7 and 8, and the quantified results of the control performance are presented in Table 3. Limited by the equipment accuracy, we set t_s in the experiment as the time when $t \geq t_s$, $|q_1 - q_{1d}| \leq 0.005$ rad. The data of e_{2s} for the proposed method and the PID method given in Table 3 is 0.000 rad because the actual angle error is less than the minimum angle error that can be measured by the encoders, which is 6.283×10^{-4} rad.

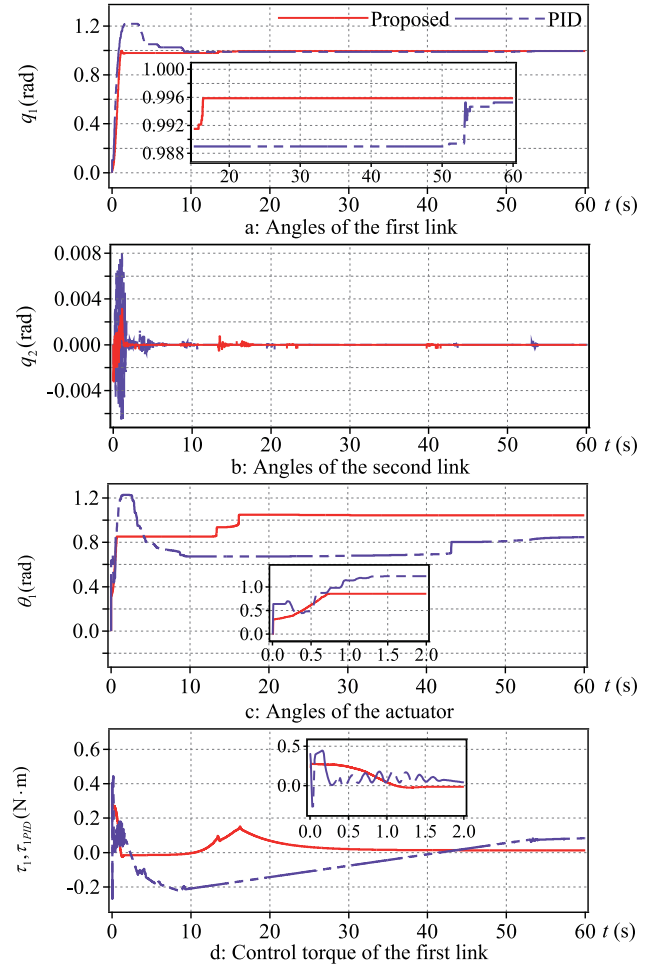


Fig. 7. Experimental results on position control of the PDFJ using the proposed method and the PID method.

Table 3

Quantified experimental results of the control performance.

Methods	t_s (s)	$\bar{h}_o$ (%)	e_{1s} (rad)	e_{2s} (rad)	τ_{1m} (N m)
Proposed	16.2	0.00	4.1×10^{-3}	0.000	0.28
PD	\	10.5	0.105	0.599	0.93
PID	57.3	21.8	4.7×10^{-3}	0.000	0.50

From Figs. 7.c and 7.d, the signals of the actuator angle θ_1 and the control torque τ_1 jump at some moments between 10 s and 20 s, mainly caused by the Coulomb friction and the effect of the NN approximators. At the moment when τ_1 exceeds the Coulomb friction, the motor angle θ_1 , which is directly controlled by τ_1 , jumps rapidly, leading to large changes in the overall velocities. As a result, the outputs of NNs change significantly, and then the torque has such a jump. It is clear that the proposed method is effective in making the PDFJ get higher positioning precision in a shorter time. The PID method makes the PDFJ exist a larger overshoot, and the PD method has unsatisfactory vibration suppression capability. Hence, the adaptive neural modified-backstepping method has better vibration suppression capability and stronger robustness.

5. Conclusion

This paper took an uncertain horizontal pendubot with double flexible joints (PDFJ) as the control object, established its dynamic model, analyzed its characteristics, and presented an adaptive neural modified-backstepping method to achieve its position control and vibration

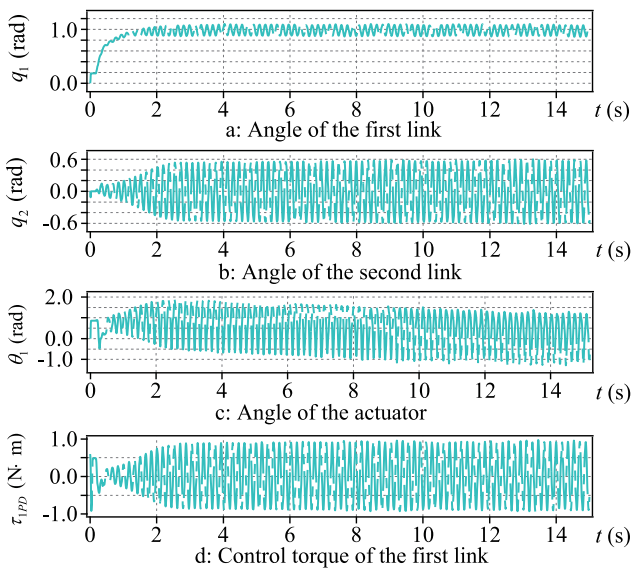


Fig. 8. Experimental results on position control of the PDFJ using the PD method.

suppression. The stability of the closed-loop system was proved based on the Lyapunov method. Simulation and experimental results verified the validity of the proposed method and showed strong robustness. This study is helpful for the research on the motion control of RAPFJs. In the future, we will extend the presented method to consider the stable boundary of the NN, which is significant work. Moreover, we will try to enhance our presented method to enable the manipulator to track a time-varying target trajectory, which aids in the execution of complex tasks with high precision. Since the PDFJ and some multi-link RAPFJs have similar physical characteristics, the proposed method can be extended to such systems as long as the approximate linear models of the systems are controllable at their equivalent points. Besides, the PDFJ is a kind of three-mass-spring system, and the presented method is applicable to such systems with one-side control input (Madonski et al., 2020; Trung & Iwasaki, 2022) as well as some underactuated multiple-mass-spring systems for which the approximate linear models of the systems are controllable at their equivalent points. Meanwhile, the applicability of the presented method in underactuated systems deserves further study.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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